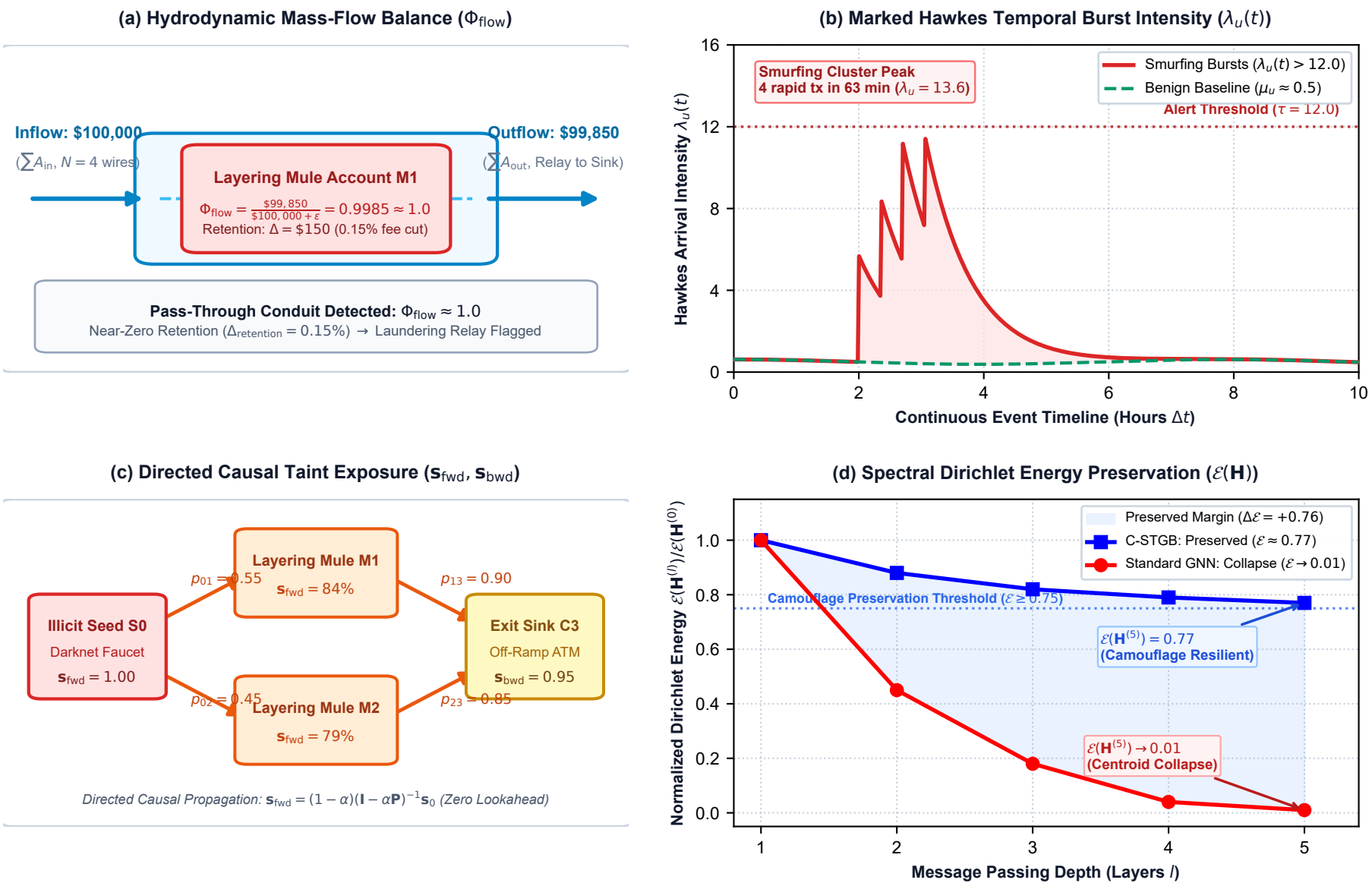

(a) Hydrodynamic Mass-Flow Balance ($\Phi_{flow}$)
Inflow: $100,000
($\sum A_{in}$, $N = 4$ wires)
Layering Mule Account M1
$\Phi_{flow} = \frac{\$99,850}{\$100,000 + \varepsilon} = 0.9985 \approx 1.0$
Retention: $\Delta = \$150$ (0.15% fee cut)
Outflow: $99,850
($\sum A_{out}$, Relay to Sink)
Pass-Through Conduit Detected: $\Phi_{flow} \approx 1.0$
Near-Zero Retention ($\Delta_{retention} = 0.15\%$) → Laundering Relay Flagged
(b) Marked Hawkes Temporal Burst Intensity ($\lambda_u(t)$)
Smurfing Cluster Peak
4 rapid tx in 63 min ($\lambda_u = 13.6$)
Smurfing Bursts ($\lambda_u(t) > 12.0$)
Benign Baseline ($\mu_u \approx 0.5$)
Alert Threshold ($\tau = 12.0$)
Hawkes Arrival Intensity $\lambda_u(t)$
Continuous Event Timeline (Hours $\Delta t$)
(c) Directed Causal Taint Exposure ($s_{fwd}$, $s_{bwd}$)
Illicit Seed S0
Darknet Faucet
$s_{fwd} = 1.00$
Layering Mule M1
$s_{fwd} = 84\%$
Layering Mule M2
$s_{fwd} = 79\%$
Exit Sink C3
Off-Ramp ATM
$s_{bwd} = 0.95$
$p_{01} = 0.55$
$p_{02} = 0.45$
$p_{13} = 0.90$
$p_{23} = 0.85$
Directed Causal Propagation: $\mathbf{s}_{fwd} = (1-\alpha)(\mathbf{I} - \alpha\mathbf{P})^{-1}\mathbf{s}_0$ (Zero Lookahead)
(d) Spectral Dirichlet Energy Preservation ($\mathcal{E}(\mathbf{H})$)
Preserved Margin ($\Delta\mathcal{E} = +0.76$)
C-STGB: Preserved ($\mathcal{E} \approx 0.77$)
Standard GNN: Collapse ($\mathcal{E} \to 0.01$)
Camouflage Preservation Threshold ($\mathcal{E} \geq 0.75$)
$\mathcal{E}(\mathbf{H}^{(5)}) = 0.77$
(Camouflage Resilient)
$\mathcal{E}(\mathbf{H}^{(5)}) \to 0.01$
(Centroid Collapse)
Normalized Dirichlet Energy $\mathcal{E}(\mathbf{H}^{(l)})/\mathcal{E}(\mathbf{H}^{(0)})$
Message Passing Depth (Layers $l$)